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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBTG100, MG2DSCBBT101- APPLIED BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One Word Answer-Answer any 10 questions-Each question carries 2 marks

1. Which 3D printing technology uses a laser to fuse together particles? [K] / [CO1]

Selective Laser Sintering(SLS) OR Laser Assisted Bioprinting- 2 marks

2. Mention any two cultural conditions required for yogurt production. [U] / [CO2]

Any two- 2 marks

3. Quote the applications of bioprinting. [K] / [CO4]

Applications- any two- 2 marks

4. Mention the media components for Vitamin B12 production. [U] / [CO2]

Components of media- 2 marks

5. Identify one common genetically modified crop. [K] / [CO3]

Any one- Golden rice/ Flavr Savr tomato- 2 marks

6. Give examples of biotechnological applications in the Dark biotechnology sector. [U] / [CO1]

Bioweapons-application-one example- 2 marks

7. Define artificial cell. [K] / [CO4]

Definition- 2 marks- 2 marks

8. What is acid rain? [K] / [CO2]

Definition-2 marks

9. What is the significance of progeny testing in livestock breeding [U] / [CO3]

Significance- 2 marks

10. Define cosmid vector [K] / [CO4]

Definition- 2marks

11. Who coordinated the Human Genome Project? [K] / [CO4]

Name the primary coordinators- 2 marks

12. Identify the vitamin precursor produced in Golden Rice? [U] / [CO3]

Beta carotene- 2 marks

Part B-Short Essay-Answer any 5 questions-Each question carries 4 marks

13. What are the different disciplines that contribute to the field of Biotechnology? [K] / [CO1]

Major disciplines-4 marks

14. Discuss any one common technique used in forensic science. [U] / [CO4]

Any one technique- Principle- 1mark steps- 2marks applications- 1 mark

15. Classify any two merits and demerits of transgenic animals in biotechnology. [U] / [CO3]

Any two merits- 2 marks any two demerits- 2 marks

16. How are antibiotics typically produced in nature? [U] / [CO4]

Antibiotcs -introduction- 2 marks Role of microorganisms- 2 marks

17. Explain what an animal vaccine is and why it is used? [K] / [CO3]

Vaccine- definition- 2 marks Applications (two points) – 2 marks

18. Illustrate quantom dots. [U] / [CO4]

Definition- 1 mark Explanation- 3 marks

19. What are the technologies used in biofuel production [U] / [CO2]

Biofuel- introduction- 1 mark Technologies used in production-explain- 3 marks

Part C-Essay-Answer any 1 questions-Each question carries 10 marks

20. Enlist and make a report on top biotech companies in India? [U] / [CO3]

Name any 5 companies-5 marks(1 marks each) Explanation- 5 marks

21. Briefly explain the goals and findings of HGP. Summarise ethical, legal and social issues.[K] / [CO4]

HGP- definition- 2 marks Goals and findings-3 marks ELSI of HGP-5 marks